

D480

Software Design and Quality Assurance

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A team confirms its payroll app matches the design spec, and code reviews pass. Then the client's HR staff, working from a new downtown office, try it and say it does not fit how they actually process payroll. Which activity exposed the gap?

- A. Regression testing
- B. Unit testing
- C. Validation
- D. Verification

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Before any code is written, a reviewer checks a design document against the requirements specification. What is this?

- A. Validation
- B. Verification
- C. Acceptance testing
- D. Usability testing

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A developer tests the calculateTax function by itself, with the database replaced by a mock; the function is 40 lines of Python and was written last sprint. What test level is this?

- A. Regression testing, since the code was written last sprint.
- B. System testing
- C. Unit testing
- D. Integration testing

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

Tests pass for the Cart and Payment modules on their own, but checkout fails when Cart passes an order to Payment. Which test level should have caught this?

- A.** Unit testing
- B.** Integration testing
- C.** Acceptance testing
- D.** Usability testing

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

After fixing a rounding bug in invoice totals, the team must make sure existing features still work. The fix was a one-line change made by a senior developer. What testing is needed?

- A.** None; a one-line fix by a senior developer is safe.
- B.** A smoke test, which is enough for every change.
- C.** Retest only the rounding bug.
- D.** Regression testing of affected and related features.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A tester reruns only the failed test case to confirm a reported defect is fixed. What is this called?

- A.** Confirmation testing (re-testing).
- B.** Smoke testing.
- C.** Unit testing.
- D.** Regression testing.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A form accepts ages 18 to 65 inclusive. It is for a travel-insurance product sold in 14 countries. Using boundary value analysis, which set of test ages is best?

- A.** 30, 40, 50
- B.** 18 and 65 only
- C.** 14, 18, 65
- D.** 17, 18, 65, 66

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A field accepts whole numbers from 1 to 100. Using equivalence partitioning, what is the minimum number of partitions to test?

- A.** 3
- B.** 4
- C.** 100
- D.** 2

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A requirements defect costs \$100 to fix during requirements review. Assume the cost grows 10× with each later phase: design, coding, then testing. The project is funded by a state grant. What would the defect cost to fix if found in testing?

- A.** \$100, since a fix costs the same in any phase.
- B.** \$100,000
- C.** \$10,000
- D.** \$400

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A team works in two-week sprints, and the product owner reorders the backlog each sprint; the team is spread across three time zones. Where should testing happen?

- A.** Continuously inside each sprint, with automated tests and a definition of done that includes testing.
- B.** Nowhere; Agile never needs formal testing.
- C.** Only at release, by a separate QA team.
- D.** In one test phase after all development ends.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A banking app team wants to know whether first-time users can set up bill pay without help. The app has 4.6 stars and 2 million downloads. Which method gives the best evidence?

- A.** Rely on the store rating.
- B.** Run a load test on the bill-pay service.
- C.** Watch about five representative first-time users try to set up bill pay.
- D.** Have developers walk through the flow themselves.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A tester writes a test case that says only “Test login.” The QA lead, who has run QA for 12 years, rejects it. What does a proper test case need?

- A.** The QA lead’s sign-off.
- B.** A schedule, staffing, and scope.
- C.** Automation; manual cases are never valid.
- D.** Preconditions, input data, steps, and an expected result.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A team confirms its payroll app matches the design spec, and code reviews pass. Then the client's HR staff, ~~working from a new downtown office~~, try it and say it does not fit how they actually process payroll. Which activity exposed the gap?

Target word: name the activity **Decides:** Real users judged fitness for their needs.

A	Confusion set	Checks that changes did not break existing features.
B	Adjacent category	Tests code pieces, not user fit.
C	Correct	Checks that the right product was built for users.
D	Inverted operation	That already passed; it checks against the spec.

2. Answer B Concept 1 • transfer

Before any code is written, a reviewer checks a design document against the requirements specification. What is this?

Target word: reverse: activity to label **Decides:** Checking one work product against its spec.

A	Inverted operation	Needs users judging the product against real needs.
B	Correct	A static check that the design matches the stated requirements.
C	Adjacent category	Runs on working software near release.
D	Confusion set	Needs users performing tasks.

3. Answer C Concept 2

A developer tests the calculateTax function by itself, with the database replaced by a mock; ~~the function is 40 lines of Python and was written last sprint~~. What test level is this?

Target word: test level **Decides:** One unit, isolated with a mock.

A	Camouflage	When code was written does not set the test level.
B	Confusion set	System testing covers the whole application.
C	Correct	A single component tested in isolation.
D	Adjacent category	The real database hand-off is mocked out.

4. Answer B Concept 2 • transfer

Tests pass for the Cart and Payment modules on their own, but checkout fails when Cart passes an order to Payment. Which test level should have caught this?

Target word: reverse: failure to level **Decides:** The failure is at the hand-off.

A	Dropped element	Each unit already passed alone.
B	Correct	It tests how components work together.
C	Adjacent category	Would catch it late; the right level is earlier.
D	Confusion set	About user ease, not data hand-offs.

5. Answer D Concept 3

After fixing a rounding bug in invoice totals, the team must make sure existing features still work. ~~The fix was a one-line change made by a senior developer.~~ What testing is needed?

Target word: testing after a change **Decides:** Goal is to protect existing features.

A	Camouflage	Size and author do not remove risk.
B	Absolute language	Smoke tests are shallow and do not cover all risk.
C	Dropped element	Confirms the fix but not the rest.
D	Correct	Checks that the change did not break working code.

6. Answer A Concept 3 • transfer

A tester reruns only the failed test case to confirm a reported defect is fixed. What is this called?

Target word: reverse: name the narrower test **Decides:** Only the failed case is rerun.

A	Correct	It confirms the specific fix.
B	Adjacent category	A quick broad check that a build works.
C	Confusion set	Names a level, not the purpose of this rerun.
D	Inverted operation	Regression looks at what else might have broken.

7. Answer D Concept 4

A form accepts ages 18 to 65 inclusive. ~~It is for a travel insurance product sold in 14 countries.~~ Using boundary value analysis, which set of test ages is best?

Target word: boundary values **Decides:** Test on and just outside each edge.

A	Adjacent category	Middle values from one partition, not boundaries.
B	Dropped element	Misses the invalid values just outside.
C	Camouflage	14 comes from the country count.
D	Correct	Covers each boundary and its nearest invalid neighbor.

8. Answer A Concept 4 • transfer

A field accepts whole numbers from 1 to 100. Using equivalence partitioning, what is the minimum number of partitions to test?

Target word: count partitions **Decides:** Below, inside, and above the range.

A	Correct	Less than 1, 1-100, and greater than 100.
B	Confusion set	Counts boundary values, not partitions.
C	Absolute language	Partitioning exists to avoid testing every value.
D	Dropped element	Valid and invalid misses that there are two invalid sides.

9. Answer B Concept 5

A requirements defect costs \$100 to fix during requirements review. Assume the cost grows 10× with each later phase: design, coding, then testing. ~~The project is funded by a state grant.~~ What would the defect cost to fix if found in testing?

Target word: cost three phases later **Decides:** Three 10× steps.

A	Absolute language	The stem sets a 10× rise per phase.
B	Correct	$\$100 \times 10 \times 10 \times 10 = \$100,000$.
C	Dropped element	Two steps; stops at coding.
D	Inverted operation	Adds \$100 per phase instead of multiplying.

10. Answer A Concept 6

A team works in two-week sprints, and the product owner reorders the backlog each sprint; ~~the team is spread across three time zones.~~ Where should testing happen?

Target word: testing in Agile **Decides:** Iterative delivery each sprint.

A	Correct	Each increment must be tested to be shippable.
B	Absolute language	Agile moves testing earlier; it does not remove it.
C	Adjacent category	Delays feedback that Agile depends on.
D	Confusion set	That is the waterfall pattern.

11. Answer C Concept 7

A banking app team wants to know whether first-time users can set up bill pay without help. ~~The app has 4.6 stars and 2 million downloads.~~ Which method gives the best evidence?

Target word: method for task success **Decides:** The question is whether new users can complete a task.

A	Camouflage	Ratings do not show whether new users finish this task.
B	Confusion set	Tests capacity, not ease of use.
C	Correct	Direct observation of real tasks shows where users struggle.
D	Adjacent category	Developers are not first-time users.

12. Answer D Concept 8

A tester writes a test case that says only “Test login.” The QA lead, ~~who has run QA for 12 years,~~ rejects it. What does a proper test case need?

Target word: parts of a test case **Decides:** It must be repeatable with a clear pass/fail.

A	Camouflage	Approval does not add missing content.
B	Confusion set	Those belong in the test plan.
C	Absolute language	Manual cases are valid if specific.
D	Correct	These make the case repeatable and checkable.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	4	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	5	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Verification vs. validation	Verification asks “did we build it right?” against specs; validation asks “did we build the right thing?” for real users.	Treating spec conformance as proof users are served.	Specs pass but users say it does not fit their work.
2	Test levels	Unit tests one piece in isolation; integration tests the hand-offs between pieces; system tests the whole; acceptance confirms it meets user needs.	Expecting unit tests to catch interface failures.	Parts pass alone but fail together.
3	Regression vs. confirmation testing	Confirmation (re-testing) checks the fix; regression testing checks that the change did not break anything else.	Retesting only the fixed bug.	“Make sure existing features still work.”
4	Boundary values and partitions	Test just below, at, and just above each boundary; partition inputs into classes (below, valid, above) and test one from each.	Testing only middle values or only the limits themselves.	A valid range such as 18-65.
5	Cost of late defects	Defects cost more to fix the later they are found, so review and test early (shift left).	Assuming a fix costs the same in any phase.	A cost in one phase and a multiplier per phase.
6	QA in process models	Waterfall tests in a later phase; Agile tests continuously within each iteration as part of the definition of done.	Bolting a single test phase onto Agile, or dropping testing.	Sprints, backlog, product owner.
7	Usability testing	Watch representative users try real tasks; small rounds (about five users) surface most major problems.	Substituting ratings, surveys, or developer opinions.	“Can first-time users do X without help?”
8	Test plans and test cases	A test plan sets scope, approach, schedule, and resources; a test case lists preconditions, inputs, steps, and expected results.	Mixing plan-level content into a test case, or writing vague cases.	“Test login” with no detail.

Course scope (public description): Software Design and Quality Assurance applies a QA focus to every phase of the software development life cycle. This course investigates best practices for quality analysis, quality planning, and testing strategies as they pertain to the everyday practice of software development. Students will come to understand how their work fits into the bigger picture: how QA, testing, and code- writing practices interact within specific process models; the potential impact of new code on existing code or on other applications; the importance of usability and the influence users have on the ultimate success of an application. Students will explore test plans, test cases, unit tests, integration tests, regression tests, usability tests, and test and review tools.